

Food Product Development

Science

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Food Product Development (A07812)

Short Title: Food Product Development
Faculty Science and Computing
Department Science
Cluster Food Science
Author EDUGGAN
Credits 15

Level: Advanced

Description of Module / Aims

This module is developed to equip students with technical knowledge, practicalities and marketing skills needed to develop and commercialise new food products, underpinning and implementing in a practical environment, theories taught in Food Innovation and Entrepreneurship. Students will develop and assess their products in terms of physicochemical and sensory properties, while also developing labelling and packaging concepts for the product/process. Global Marketing Communications Strategies will explore current and emerging international communications approaches to develop an integrated communications approach. The course will challenge students to develop creative strategies on international issues in advertising, public relations, diverse media channels, personal selling, word-of-mouth marketing, event management, negotiating and sales promotion. The course includes a study of innovative digital marketing techniques.

Programmes

	BA (Hons) in Culinary Arts (WD_HCULA_B)	4 / 8 / E
SCIE-0065	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	4 / 8 / M
SCIE-0065	BSc (Hons) in Food Science and Innovation (WD_SFDSC_B)	4 / 2 / M

Indicative Content

- Food product development: implementation of new food product development process; food prototyping & development evaluation; discovery, scoping and building a business case; development, testing and validation of new food product prototypes; post launch review and evaluation of metrics; barriers/gates encountered during phased stages of new food product development
- Sensory testing & analysis: evaluation of food physical characteristics, i.e. colour, texture and viscosity in beverages; sensory evaluation of food - taste panels
- Nutritional declaration, packaging & labelling: design food label based on legal legislation for new product; suitable packaging systems for new food products
- International marketing and digital marketing: developing an integrated global communication strategy; implementing the elements of the marketing mix; digital Marketing; digital media; measuring digital marketing campaigns

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Appraise, reflect and maintain detailed records of all stages and activities associated with new food product development.
2. Interpret the regulations involved in food, labelling and nutrition declaration of food products.
3. Critique the various types of food packaging available.
4. Develop a food product/ process and present to industry stakeholders.
5. Evaluate the new food product in terms of objective and sensory analysis.
6. Determine differences between standardisation and adaptation in communication strategies across cultural and international borders.
7. Assess appropriate market entry strategies.
8. Critique international food marketing case studies.
9. Recommend cross-cultural marketing strategies.
10. Create effective digital marketing tactics.

Learning and Teaching Methods

- Lectures will provide the theory for varied aspects of the module.
- Kitchen-based practical classes will enable students to manufacture and develop new products.
- Lab-based practical classes will enable students to perform food analysis experiments to determine physico-chemical properties of the developed product.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	25%	6,7,8,9
Continuous Assessment	75%	
<i>Project</i>	50%	1,2,3,4,5,10
<i>Presentation</i>	25%	4

Assessment Criteria

- <40%:** Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. May have some problems with critical thinking and problem solving. Adequate practical skills.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Good practical skills.
- 60%-69%:** Demonstrate skilled and sound knowledge of the subject matter. Show ability to analyse and logically discuss in an effective way. Good critical thinking and a high level of competence and efficiency in theoretical and practical areas of this subject.
- 70%-100%:** All of the above to an excellent level. Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	120	
Independent Learning	249	

Essential Material

- "FSAI." www.fsai.ie/legislation/food_legislation/food_information_fic/food_information-fic.html
- Beckley, J., M. Foley, E. Topp, J.C. Haung and W. Prinyawiwatkul. *Accelerating New Food Product Design and Development*. Ames, Iowa: Blackwell Publishing, 2007.
- Brody, A.L. and J.B. Lord. *Developing New Food Products for a Changing Marketplace*. London: CRC Press, 1999.
- Cateora, P.R. and P.N. Ghauri. *International Marketing European Edition*. London: McGraw-Hill, 2005.
- Clarke, G. and I. Wilson. *International Marketing*. Maidenhead, Berkshire, UK: McGraw-Hill, 2009.
- Coburn, J. *How to Win Business on the Internet*. DVD. 2009. Ireland: praxisnow.ie.
- Fuller, G.W. *New Food Product Development: From Concept to Marketplace*. London: CRC Press, 2005.
- Hollensen, S. *Global Marketing: A Decision Orientated Approach*. 4th ed.. Harlow, UK: Prentice Hall, 2007.

Supplementary Material

- Crawford, C.M. *New Products Management*. Boston, USA: McGraw-Hill, 1997.
- Hart, S. *New Product Development*. London: Dryden Press, 1996.
- McGee, H. *McGee on Food and Cooking: an encyclopedia of kitchen science, history and culture*. London: Hodder and Stoughton, 2004.
- Patrick, J. *How to Develop Successful New Products*. Chicago: NTC Business, 1997.
- Thomas, R.J. *New Product Development: Managing and Forecasting for Strategic Success*. New York: Wiley, 1993.
- Turban, E. *Electronic Commerce: A Managerial Perspective*. London: Prentice Hall, 2008.

Requested Resources

- HOSPITALITY: Kitchen
- Room Type: Chemistry Lab